TREES
and
RANDOM FORESTS

TREES

From “Decision Tree Algorithm Explained with Examples” (08/2021)
https://www.mygreatlearning.com/blog/decision-tree-algorithm/

Assumptions:

1- Discretization of continuous variables is required

2- The data taken for training should be wholly considered as root

3- Distribution of records is done in a recursive manner on the basis of attribute values.

TREES- Processes involved in Decision Making

From “Decision Tree Algorithm Explained with Examples” (08/2021)
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1-Entropy

2-Information Gain:

Information Gain = Entropy of Parent – sum (weighted % * Entropy of Child)

Weighted % = Number of observations in particular child/sum
(observations in all child nodes)

3-Gini - “grau de variedade” em cada nó

(busca-se variedade zero)

4-Reduction in Variance

-tree works for regression and the output is continuous

-criteria of splitting are selected only when the variance is reduced to minimum

TREES- ALGORITHMS

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ID3 - Entropy / Information Gain

C4.5 - **NORMALIZED** information gain

CART

-greedy algorithm to reduce cost function
-for classification, Gini index cost function
-for regression, sum squared error

TREES – RANDOM FORESTS

From “An intuitive explanation of random forests” (08/2021)
https://towardsdatascience.com/an-intuitive-explanation-of-random-forests-109b04bca343
“The Intuition behind Random Forest”(08/2021) – https://www.kaggle.com/general/191828

WHY ?

To foster PRECISION (decrease VARIANCE + maintain BIAS)

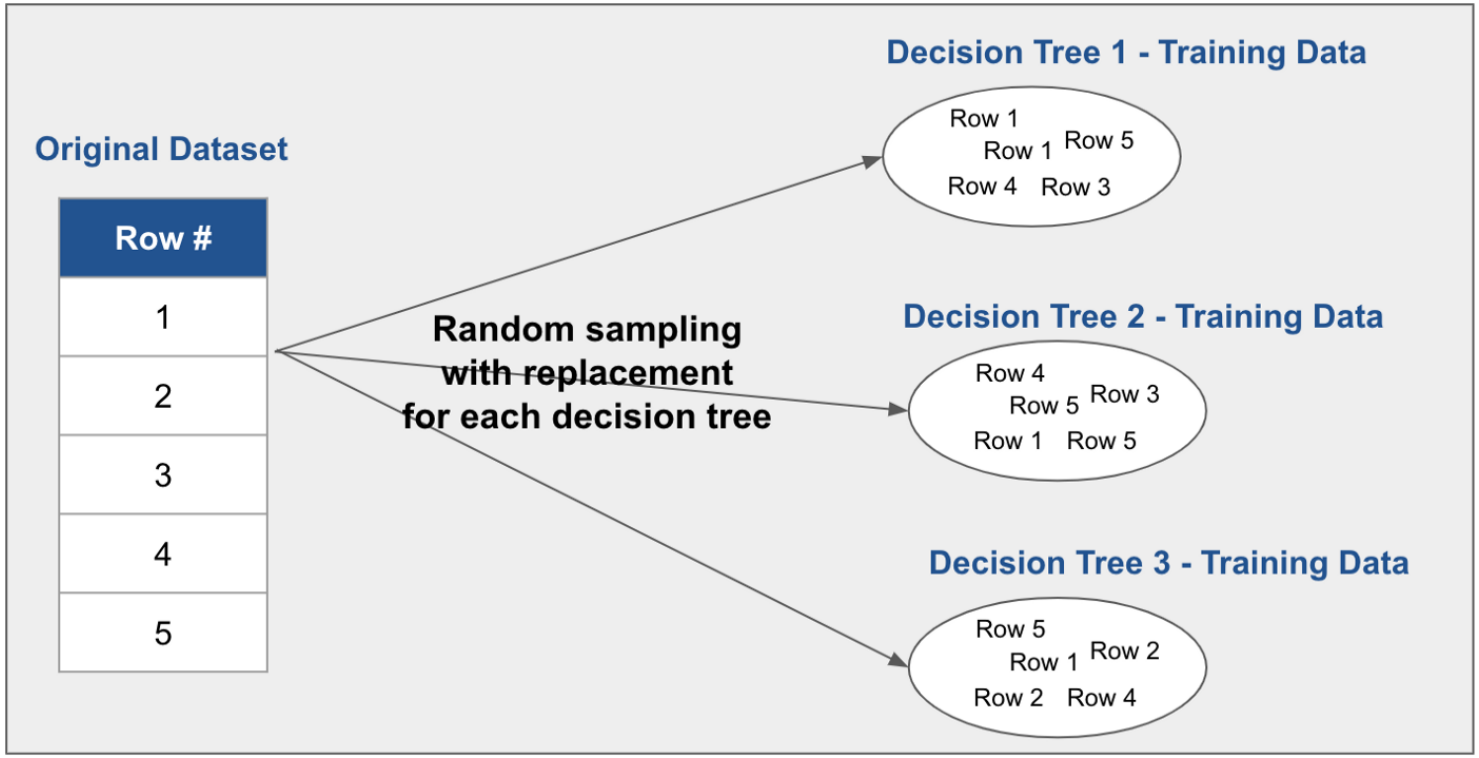
→ FLEXIBILITY to unseen examples

BIAS and VARIANCE - (STATQUEST)
https://www.youtube.com/watch?v=EuBBz3bI-aA

RANDOM FORESTS: BUILDING, USING, EVALUATING - (STATQUEST)
https://www.youtube.com/watch?v=J4Wdy0Wc_xQ

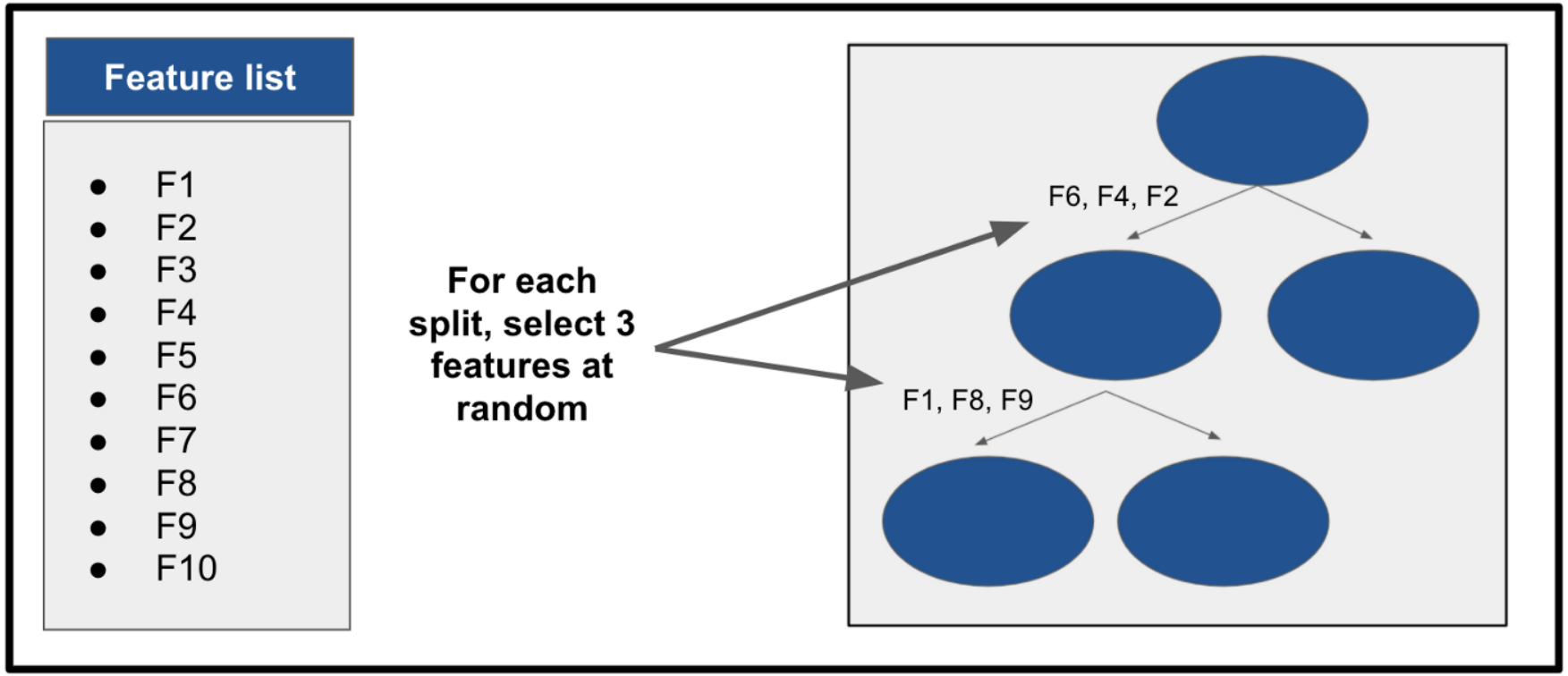
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INTUITION

Random forest algorithm encourages **DIVERSITY**:

- each decision tree is trained on a DIFFERENT SUBSET of the data (each decision tree “knows” a different subset of the data)
- selecting a RANDOM SUBSET(of size M) of the full list of features at every split

This process generates trees that have been exposed to DIFFERENT INFORMATION, which are then AGGREGATED together

TREES – RANDOM FORESTS

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HYPERPARAMETERS

1-MAX FEATURES: how many features are available at each split
(common default is the **square root** of the
total number of features)

2-COST FUNCTION: method used to identify the best split
in each tree — gini impurity, feature importance, etc.

3-NUMBER OF ESTIMATORS: how many trees are in the forest.

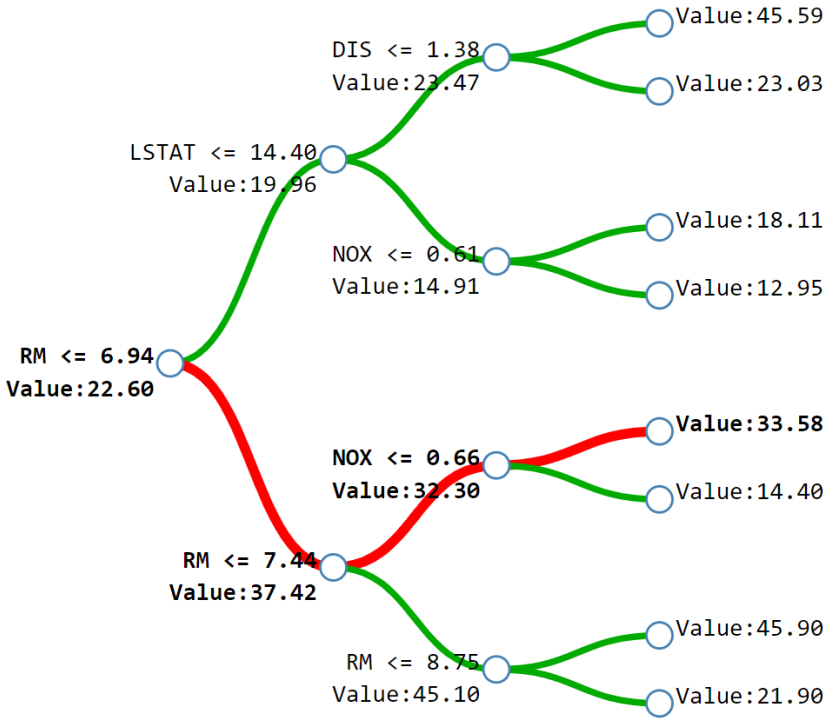
4-MAX DEPTH: how far from the root node can each tree grow.

5-MINIMUM SAMPLES FOR A SPLIT: number of records that must be
in each node to proceed with another split

RANDOM FORESTS - INTERPRETABILIDADE

From “Interpreting random forests” (08/2021)
http://blog.datadive.net/interpreting-random-forests/

-breakdown of the prediction, in terms of value changes along the path
-together with feature names that “caused” every value change



INTUITION:
- there is a more “operational” way to define the prediction: through the sequence of **REGIONS of the FEATURE SPACE** that correspond to **each node/decision** in the tree.
- for each decision that a tree (or a forest) makes there is a path (or paths) from the Root of the tree to the leaf, consisting of a series of decisions, guarded by a particular feature
- each decision either adds or subtracts from the value given in the parent node
-the prediction can be defined as the sum of the feature contributions + the “bias” (i.e. the mean given by the topmost region that covers the entire training set).

Prediction: **33.58** ≈ **22.60** (trainset mean) + **14.82**(gain from RM) - **5.12**(loss from RM) + **1.28**(gain from NOX)

RM	LSTAT	NOX	DIST	
3.1	4.5	0.54	2.6	Predict
6.5	16.1	0.12	2.2	Predict
7.1	10.5	0.31	1.8	Predict

CLASSICAL ML BOOK: https://hastie.su.domains/ElemStatLearn/